

Task 1: Business Requirements Analysis

1. Company Overview & Organisational Structure

Ubuntu Innovations (Pty) Ltd is a rapidly growing technology startup based in Cape Town, South Africa. The company is relocating to a new office and requires a fully documented IT infrastructure built from scratch across its 25-person workforce.

Organisational Structure	Company Context
Department Executive Management Finance Human Resources Sales & Marketing Software Development TOTAL: 25 employees	Industry: Information Technology / Software Development Stage: Early-growth startup Location: Cape Town, Western Cape, South Africa Office: New premises — greenfield IT deployment IT Maturity: Low — no existing legacy infrastructure Operating Hours: Mon–Fri, 08:00–17:00 SAST Users: 25 full-time employees across 5 departments

2. Business Requirements Summary

The following requirements were identified across all departments and must be addressed through the IT infrastructure design, security policies, and support procedures.

#	Category	Requirement	Priority
1	Connectivity	Wired & wireless LAN for all 25 users; business-grade firewall router	Critical
2	User Management	Individual accounts with dept-based security groups & RBAC	Critical
3	File Storage	NAS with departmental folder structure and access permissions	Critical
4	Printing	Two shared network laser printers accessible across the LAN	Medium
5	Data Security	Antivirus, device encryption (laptops), MFA, patch management	Critical
6	Backup & Recovery	Daily incremental, weekly full, monthly archive; cloud offsite copy	High
7	Applications	Microsoft 365 for all staff; dept-specific software per section 3	Critical
8	Power Protection	Two 1500VA UPS units protecting core networking equipment	High
9	Documentation	Troubleshooting guide and IT support procedures for staff	Medium

3. Business Application Requirements by Department

Department	Required Software	Notes
All Staff (25)	Microsoft 365, Antivirus (Bitdefender), Zoom, Web Browser	Core productivity and security baseline for every device
Executive Mgmt (3)	Power BI, Asana, BitLocker encryption	Laptops must be fully encrypted; mobility-focused
Finance (4)	Sage Business Cloud, Advanced Excel, Seclore DRM	SARS-compliant; payroll data requires strict access control
Human Resources (3)	SimplePay, SharePoint (M365), DocuSign	Employee records need privacy controls and digital signing
Sales & Marketing (7)	HubSpot CRM, Canva Pro, Hootsuite, Google Analytics	Cloud-dependent; stable internet access is critical
Software Dev (8)	VS Code, Git, GitHub, Docker, Ubuntu Linux 22.04	Developer workstations run Linux; elevated permissions needed

4. Operational Challenges

Six key operational challenges have been identified that the IT infrastructure design must address from the outset.

Challenge	Description	Challenge	Description
Greenfield Deployment	All infrastructure must be built from scratch before operations begin.	Data Segregation	Finance and HR hold sensitive data requiring strict access controls.
Load Shedding	SA power outages risk downtime and data loss without UPS protection.	Cybersecurity Threats	Phishing and ransomware are high-probability risks requiring MFA and AV.
Scalability	Infrastructure must accommodate projected staff growth without a rebuild.	Software Licensing	25 devices require tracked, compliant licences across all applications.

5. Data Security & Backup Requirements

Data Security Controls	Backup & Recovery Schedule		
Antivirus & endpoint protection on all 25 devices BitLocker encryption on all 5 executive laptops Multi-factor authentication (MFA) for all accounts Security patching within 72hrs (critical) / 14 days (standard) Written Acceptable Use Policy issued to all	Frequency	Type	Retention
	Daily	Incremental	7 days rolling
	Weekly	Full backup	4 weeks
	Monthly	Archive	12 months
	Continuous	Cloud offsite	Real-time

staff Physical access controls for server room and network rack	RTO: < 4 hours RPO: < 24 hours NAS with RAID-5 as primary backup target
--	---

6. Connectivity & Network Requirements

Wired & Wireless Network	IP Addressing Plan (Preview)	
Gigabit Ethernet for all 25 desktop workstations 24-port managed switch for internal LAN 2× Wi-Fi 6 dual-band access points for wireless coverage Business-grade firewall router (NAT, VPN, traffic filtering) VLAN segmentation: servers, users, printers, Wi-Fi 2× 1500VA UPS units for core equipment power protection	Device Type	Subnet
	Servers / NAS	192.168.10.0/24
	User Devices	192.168.20.0/24
	Printers	192.168.30.0/24
	Wi-Fi Clients	192.168.40.0/24

7. Key Recommendations

#	Recommendation
1	Deploy a VLAN-segmented network from day one to ensure security and scalability as the company grows.
2	Implement centralised identity management (Active Directory or equivalent) for user accounts and group policy enforcement.
3	Enforce a zero-trust security baseline from the outset: MFA on all accounts, endpoint AV, encrypted laptops, and a signed AUP.
4	Select all hardware with a minimum 3-year South African vendor warranty to reduce support risk and downtime.
5	Design the backup strategy to satisfy both on-premises (NAS/RAID-5) and offsite (cloud replication) requirements.
6	Document all configurations, procedures, and troubleshooting guides immediately to reduce single-point-of-knowledge risk.
7	Plan a 30% growth buffer in IP addressing, switch port capacity, and NAS storage to accommodate future hiring.

Task 2: Hardware Inventory

Hardware Inventory Table

The table below details all recommended hardware assets for Ubuntu Innovations (Pty) Ltd. All costs are estimated in South African Rand (ZAR).

Asset	Qty	Suggested Specification	Unit Cost (ZAR)	Total Cost (ZAR)	Business Justification	Warranty
Desktop Computers	20	Intel Core i5, 16GB RAM, 512GB SSD, Windows 11 Pro	R12,000	R240,000	General office use across all 5 departments	3-year on-site
Laptops	5	Intel Core i7, 16GB RAM, 512GB SSD, Windows 11 Pro + BitLocker	R18,000	R90,000	Mobility for Executive Management team	3-year on-site
Router / Firewall	1	Business-grade firewall router (e.g. Cisco RV340 / Fortinet 40F)	R8,500	R8,500	Secure internet access, NAT, VPN and traffic filtering	3-year vendor support
Managed Switch	1	24-port Gigabit managed switch (e.g. Cisco SG350-28)	R4,500	R4,500	Internal LAN for all 25 wired devices plus spare ports	3-year vendor support
Wireless Access Points	2	Dual-band Wi-Fi 6 AP (e.g. Ubiquiti UniFi U6-LR)	R3,200	R6,400	Full wireless coverage across the office floor	2-year manufacturer
Network Printers	2	Monochrome laser printer with network card (e.g. HP LaserJet Pro M404dn)	R5,500	R11,000	Shared departmental printing — low cost per page	1-year; service contract optional
NAS Device	1	8TB RAID-5 NAS (e.g. Synology DS423+ with 4×4TB WD Red drives)	R14,000	R14,000	Centralised file storage and daily backup target	3-year manufacturer

UPS Units	2	1500VA UPS (e.g. APC Smart-UPS 1500VA LCD)	R3,800	R7,600	Power protection for network rack during load shedding	2-year manufacturer
TOTAL ESTIMATED BUDGET				R382,000		